

Problem Set 3

1. Ideal sinc reconstruction

Let the discrete-time sequence be $x[n] = \delta[n] + 2\delta[n - 1]$, obtained from a valid bandlimited sampling process with sample period T .

- Write the ideal continuous-time reconstructed signal $x_r(t)$ using normalized $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.
- Verify $x_r(0) = 1$ and $x_r(T) = 2$.
- Compute $x_r(T/2)$.

2. Zero-order hold frequency response

A DAC uses a zero-order hold $h_{ZOH}(t) = u(t) - u(t - T)$.

- Derive $H_{ZOH}(j\Omega)$ by direct integration.
- Separate the result into a linear-phase term and a sinc-like magnitude term.
- Compute the normalized magnitude $|H_{ZOH}|/T$ in dB at $\omega = \Omega T = 0.5\pi$ and at $\omega = \pi$.

3. Uniform quantization and SQNR

A uniform B-bit quantizer spans approximately -1 V to $+1$ V. Assume the high-resolution uniform-error model when requested.

- For $B = 8$, find the quantization step Δ .
- Find the modeled error variance $\sigma_e^2 = \Delta^2/12$ and RMS quantization error.
- Estimate the SQNR of a full-scale 8-bit sine.
- Estimate the full-scale sine SQNR for 16 bits.
- Explain why “16-bit audio is about 96 dB” and “16-bit full-scale sine SQNR is about 98 dB” are not necessarily contradictory

4. Finite sequence z-transform

Let $x[n] = 2\delta[n + 1] - \delta[n] + 3\delta[n - 2]$. Find $X(z)$, state the ROC, and determine whether the DTFT exists.

5. Right-sided exponential and convergence

For $x[n] = (1.2)^n u[n]$, derive $X(z)$ from the geometric series, state the ROC, identify the pole, and determine whether its DTFT exists.

6. Same algebra, different ROC

Find the inverse z-transform of $X(z) = 1/(1 - 0.75z^{-1})$ for

a) $|z| > 0.75$ and

b) $|z| < 0.75$.

Explain in one sentence why the two answers can have the same rational expression.

7. Partial fractions with two ROCs (ECEN 5632 Problem)

Let $X(z) = 1/[(1 - 0.5z^{-1})(1 - 0.8z^{-1})]$. Use partial fractions and find $x[n]$ for

a) $|z| > 0.8$ and

b) $0.5 < |z| < 0.8$.